



SIGN EASE

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Abstract

- SignEase is a web-based sign-language learning and translation prototype. It provides a sign dictionary, learning modules, camera access and speech output
- Brings sign-learning and demo translation features together in one accessible platform.

Problem Statement & Objectives

- Problem: Communication can be difficult when people do not understand sign language, and learning resources may be scattered.
- Objective 1:Demo sign-to-text translation
- Objective 2:Sign dictionary and lessons
- Objective 3:Speech output and saved history

Existing vs Proposed System

Feature	Existing System	Proposed System
Learning	scattered resources	Dictionary and lessons
Translation	sign language expert	sign-to-text workflow
Accessibility	seperate tools	one browser based interface
saved activity	maynot be organized together	History and favorites
Recognition	Limited basic tools	ML recognition planned for future

Feasibility Study

- Technical:Built using HTML, CSS, JavaScript and browser APIs
- Economic:Uses open-source web technologies with no special hardware requirement.
- Operational: Simple interface with separate modules for easy navigation.
- Schedule: Prototype designed for completion within the semester.

Results & Testing

Results: Demo translation, dictionary, lessons and speech output.Testing: Navigation, camera and speech tested.Limitation: Limited sign dictionary

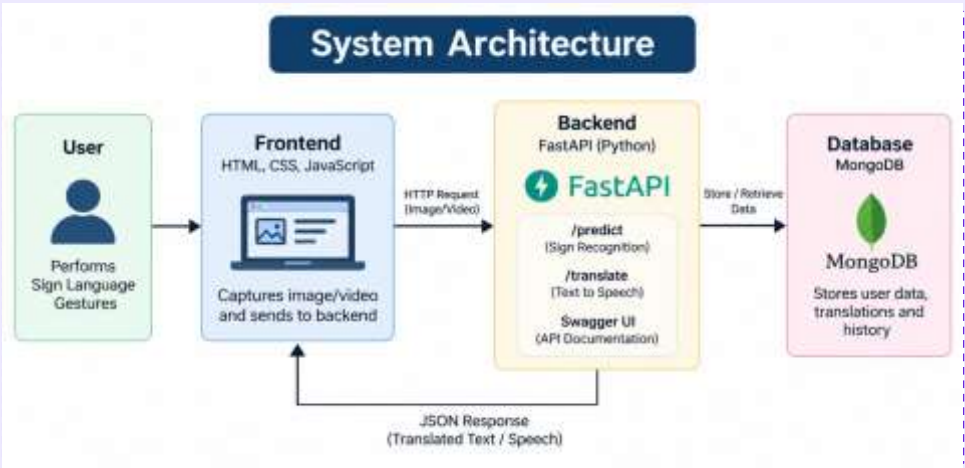
Conclusion

SignEase brings sign-language learning and demo translation into one browser-based interface. It supports text and speech output.

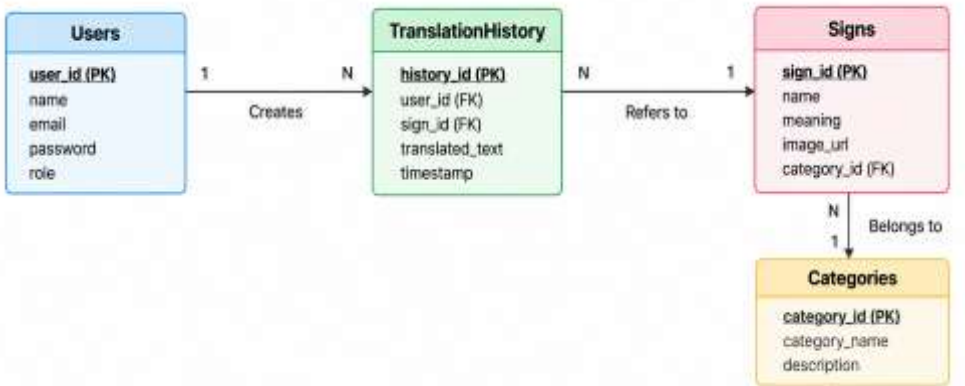
Tools & Technologies

- Front-End: HTML,CSS,JavaScript
- Back-End: Fast API , Swagger
- Database: MongoDB
- Tools: VS Code , chrome Browser

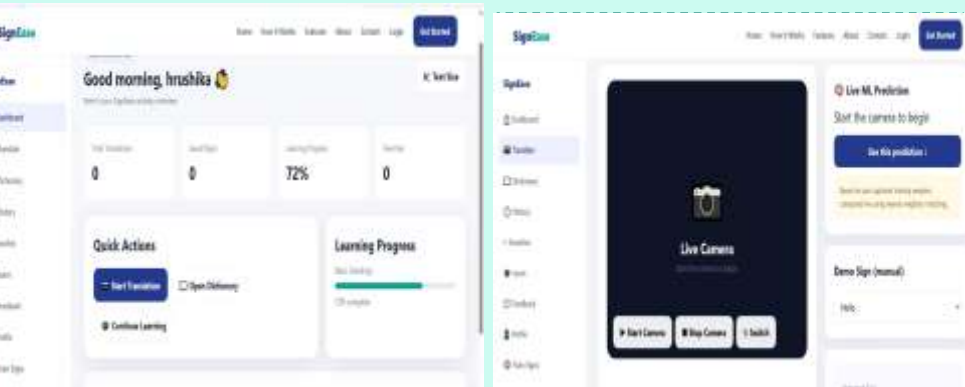
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Expand the sign dictionary and learning content.

